

HANOI UNIVERSITY OF SCIENCE AND
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SCHOOL OF INFORMATION AND
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OBJECT-ORIENTED
PROGRAMMING
PROJECT REPORT
LAB05 ASSIGNMENT

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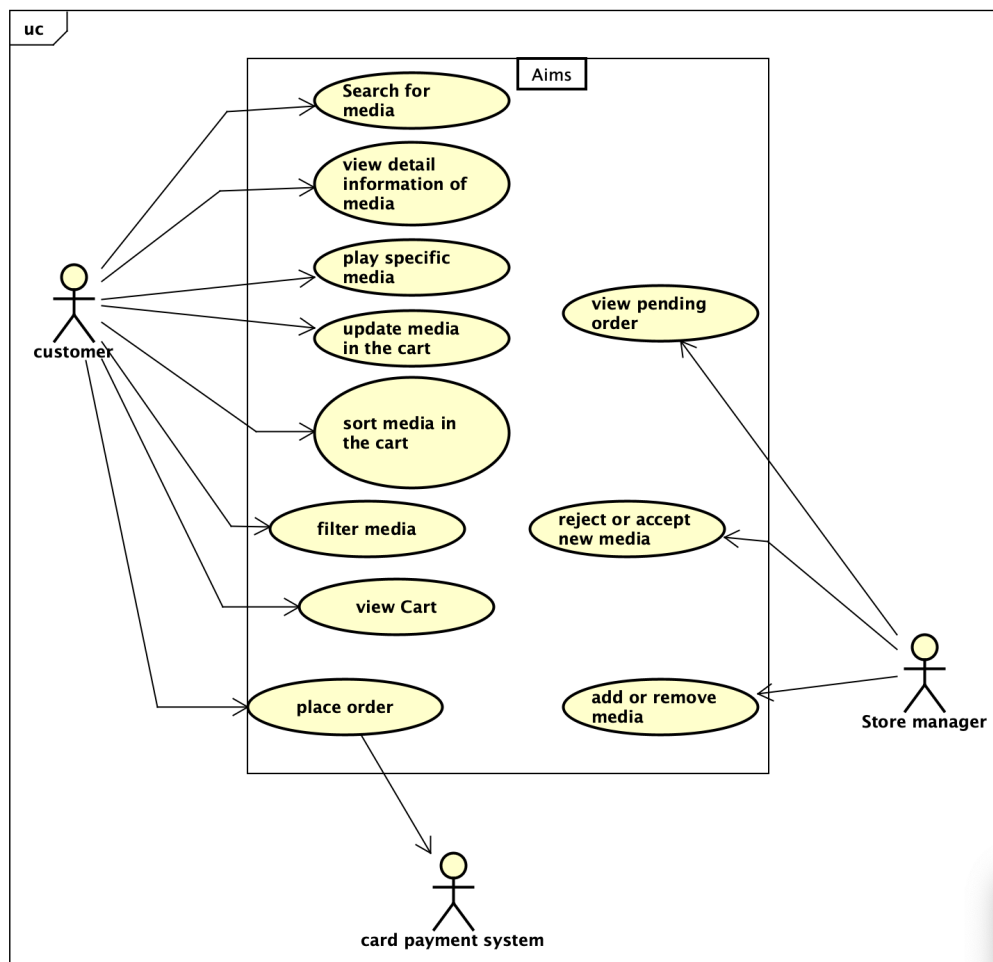
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1 UseCase Diagram

1.1 Introduction about system and objective

This document outlines the main functionalities of the AIMS System, focusing on the interactions between the two primary user groups: Customer and Store Manager, and the external Payment System integration.

1.2 Diagram



1.3 Actor

- Customer: User who use the service of the system to search and buy media
- Store Manager: People who can manage the store, update the item of store
- Payment System: The outer system to place order.

1.4 UseCase

Use Case	Actor	Description
Browse/Search Media	Customer	Search or browse media products within the store.
View Media Details	Customer	View detailed information about a media item (DVD, CD, Book).
Play Media	Customer	Play media content, e.g., CD/DVD (feature not yet supported).
Add/Remove to Cart	Customer	Add or remove media items from the shopping cart.
Sort/Filter Cart	Customer	Sort and filter media items within the shopping cart.
Place Order	Customer	Place the selected media items in the cart as an order.
Make Payment	Customer/Payment	Pay for the order (linked with an external payment system).
Login	Store Manager	Log in to the management system.
View/Approve/Reject Order	Store Manager	Manage, approve, or reject customer orders.
Add/Remove Media	Store Manager	Add new or remove media products from the inventory.

2 Class Diagram

2.1 Objective

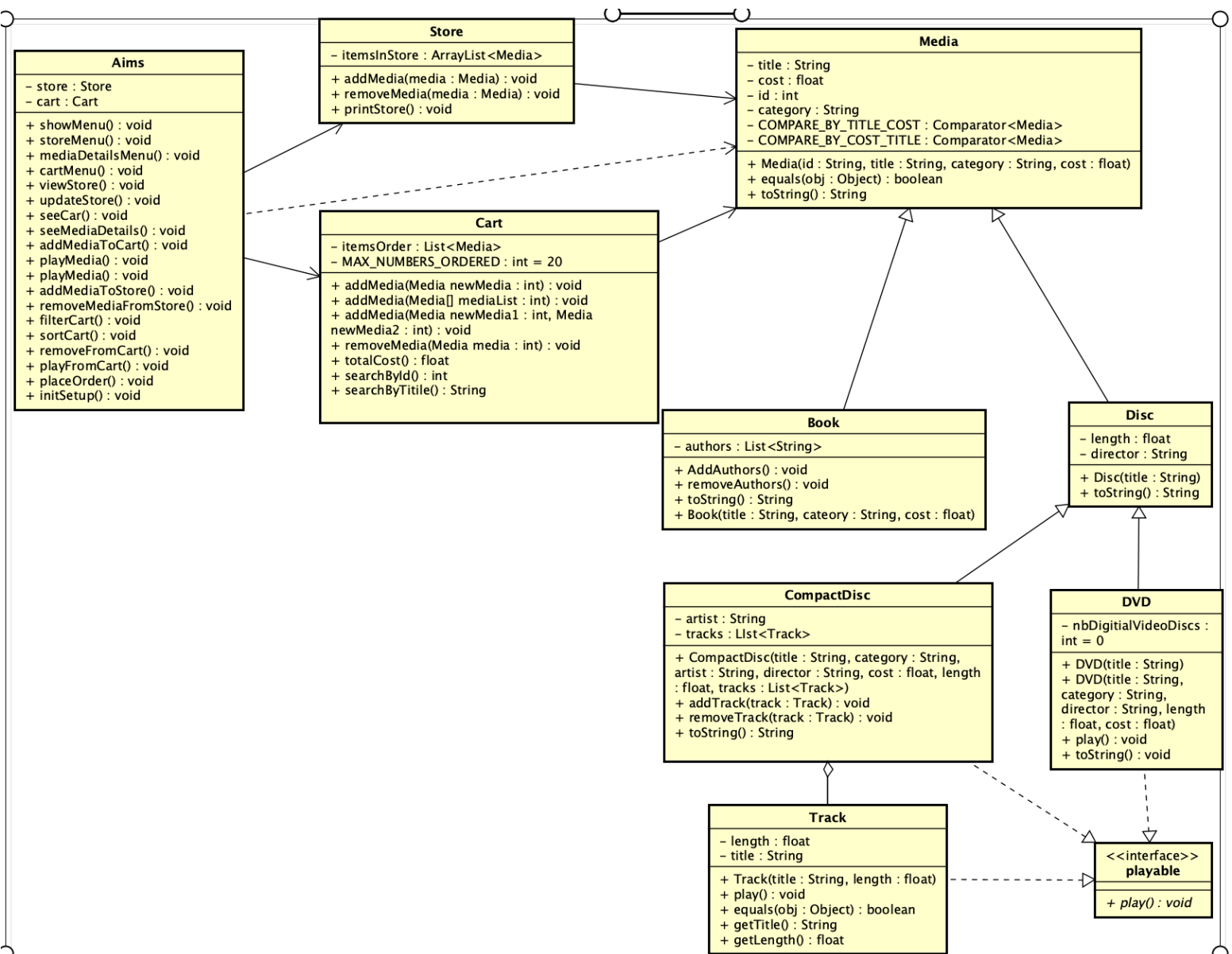
Class diagram illustrates the main object classes, their attributes, methods, and the relationships between these classes within the AIMS system. This diagram will serve as the foundation for building the Object-Oriented Programming (OOP) architecture of the system

2.2 Genral Structure

Class	Function
Media	Parent class for all media types; contains common information such as id, title, category, and cost.
Disc	Intermediate class for disc types; adds director and length attributes.
DigitalVideoDisc	Represents a DVD; capable of being played.
CompactDisc	Represents a CD; contains a list of tracks and is playable.
Track	Represents a song belonging to a CD; capable of being played.
Book	Represents a book; contains a list of authors.
Playable	Interface defining the play() method for playable media types.
Cart	Manages the customer's shopping cart; allows adding, removing, sorting, and filtering media.
Store	Manages the store's media inventory; supports searching, adding, and removing media.

Bảng 1: Brief description of classes

2.3 Diagram



Hình 1: UML Class Diagram for the AIMS Media Management System. This diagram illustrates the relationships between Media, Cart, Store, and the Playable interface.

2.4 Discussion

Main Packages *The system is organized into the following main packages:* `media`, `cart`, `store`, `main`

- **Abstract Base Class:** The `Media` class is the abstract parent class for all media objects, including `DVD`, `CompactDisc`, and `Book`.
- **Intermediate Class:** The `Disc` class acts as an intermediate abstract class between the abstract `Media` class and the concrete disc types (`DVD`, `CompactDisc`).
- The `Playable` interface is implemented by media classes that can be played, such as `DVD`, `CompactDisc`, and `Track`.
- **Store:** The `Store` class manages the list of media objects currently available in the inventory.
- **Cart:** The `Cart` class manages the list of media objects that the customer intends to purchase.

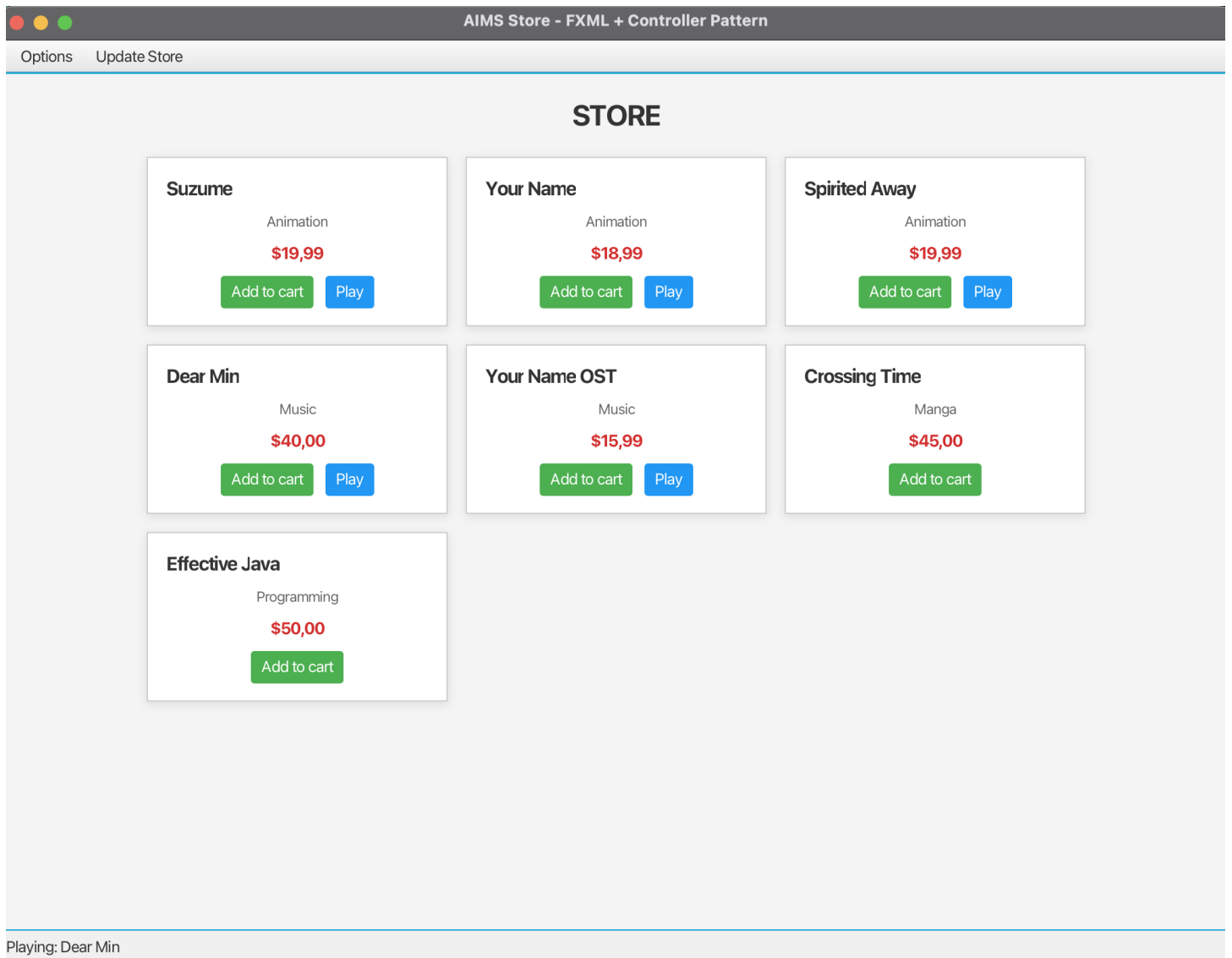
3 Use Application ---

3.1 Store Screen

The Store screen displays a summary of the media inventory. It offers features like playing CDs and DVDs, and adding any media item to the cart.

3.2 Show information of media

When a user selects a media item, a detailed view pops up. For instance, clicking the 'Dear Min' CD reveals comprehensive information such as the artist, the track count, the length of every track, and the price. Beside that, we also have "add to cart" and "play" functions



Hình 2: AIMS Store

3.3 Play function

The screen will show which tracks can be played successfully

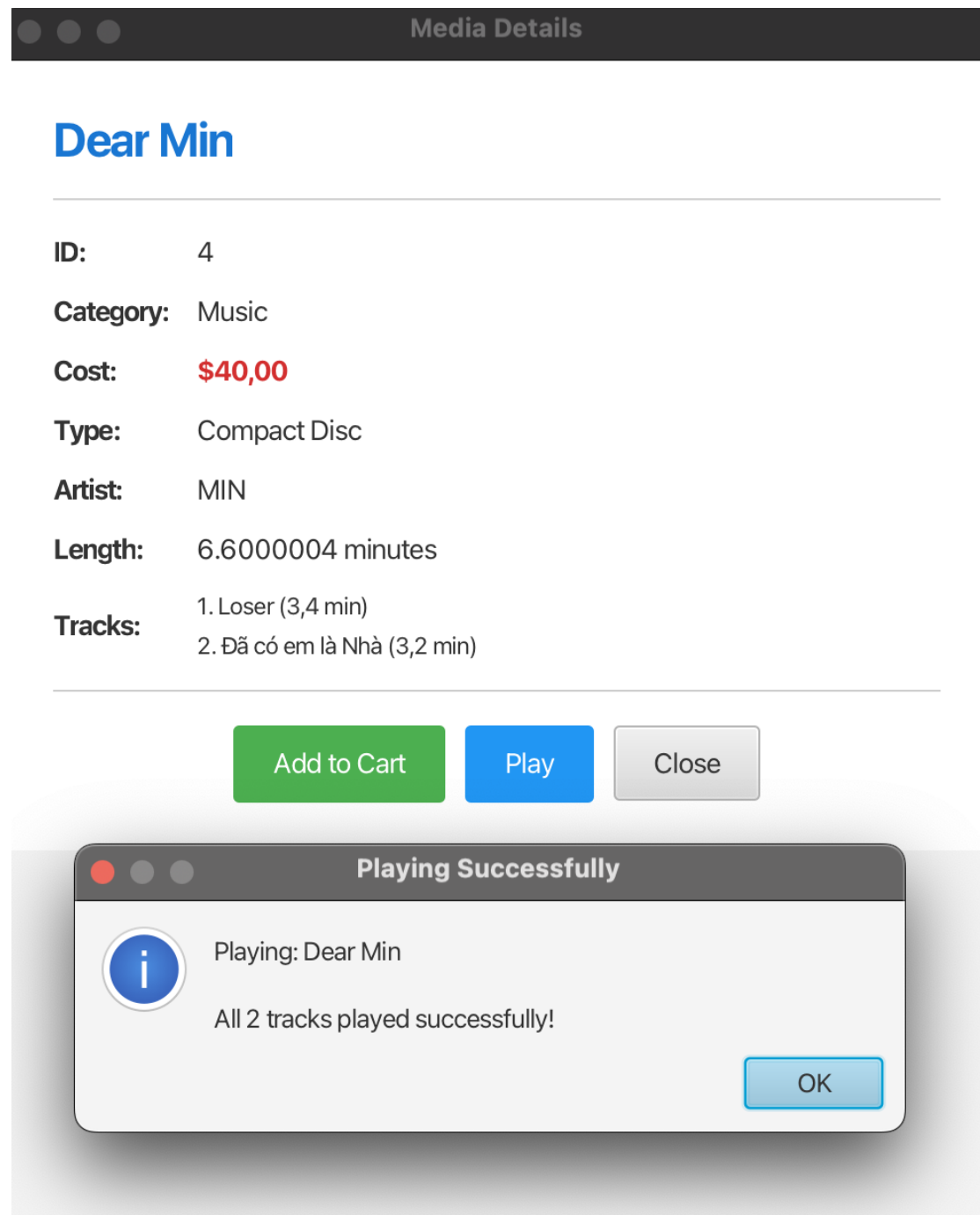
Dear Min

ID: 4
Category: Music
Cost: **\$40,00**
Type: Compact Disc
Artist: MIN
Length: 6.6000004 minutes
Tracks:
1. Loser (3,4 min)
2. Đã có em là Nhà (3,2 min)

[Add to Cart](#)[Play](#)[Close](#)

Hình 3: Information of media

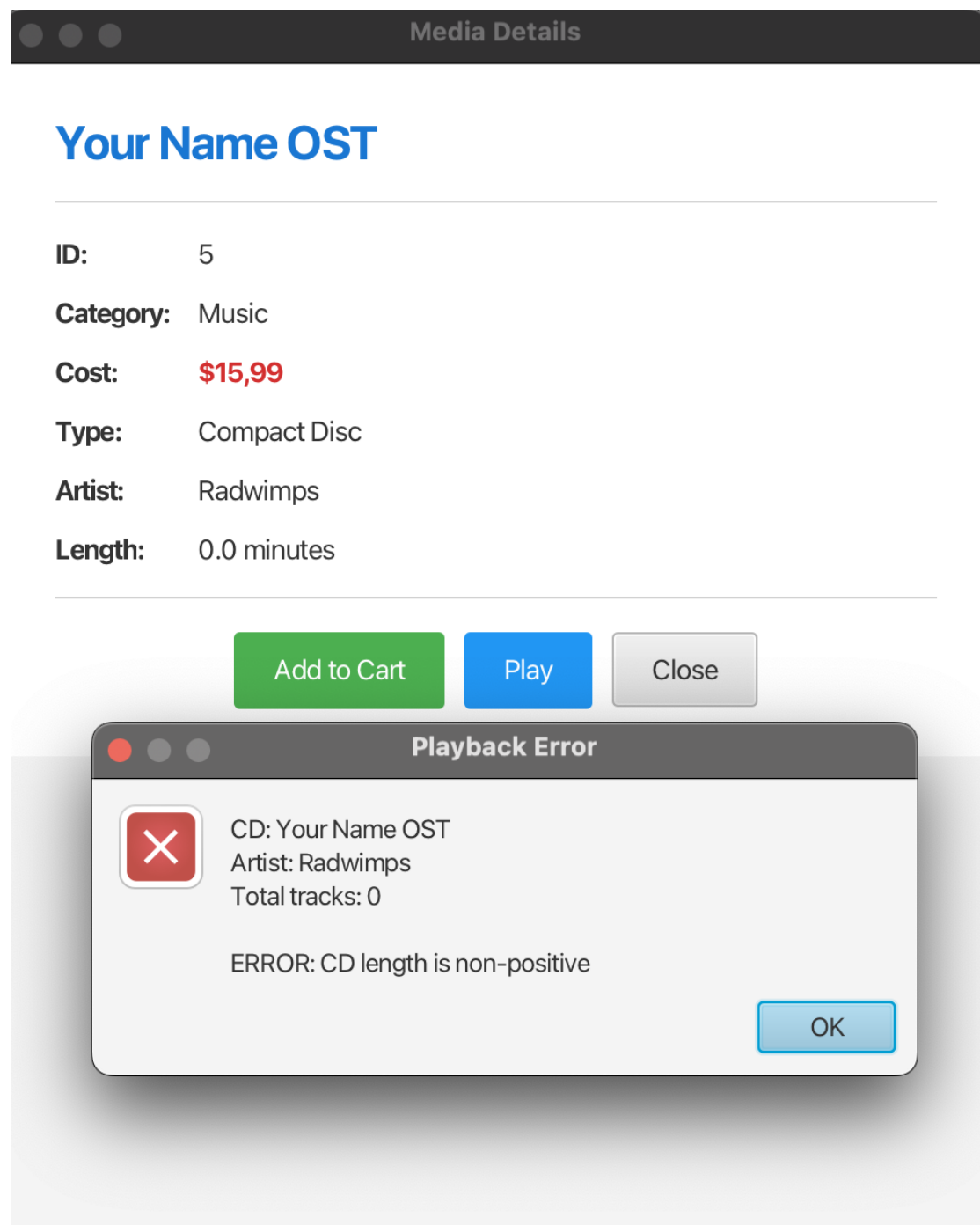
Otherwise, the screen will show the error message, when it has an exception.
For example, a CD has the non-positive length.



Hình 4: Play function

3.4 Add to Cart

When you find a media item that interests you, you can consider adding it to your cart via the '**Add to Cart**' button on the screen. We also add some other media to the cart, and view them in the cart.

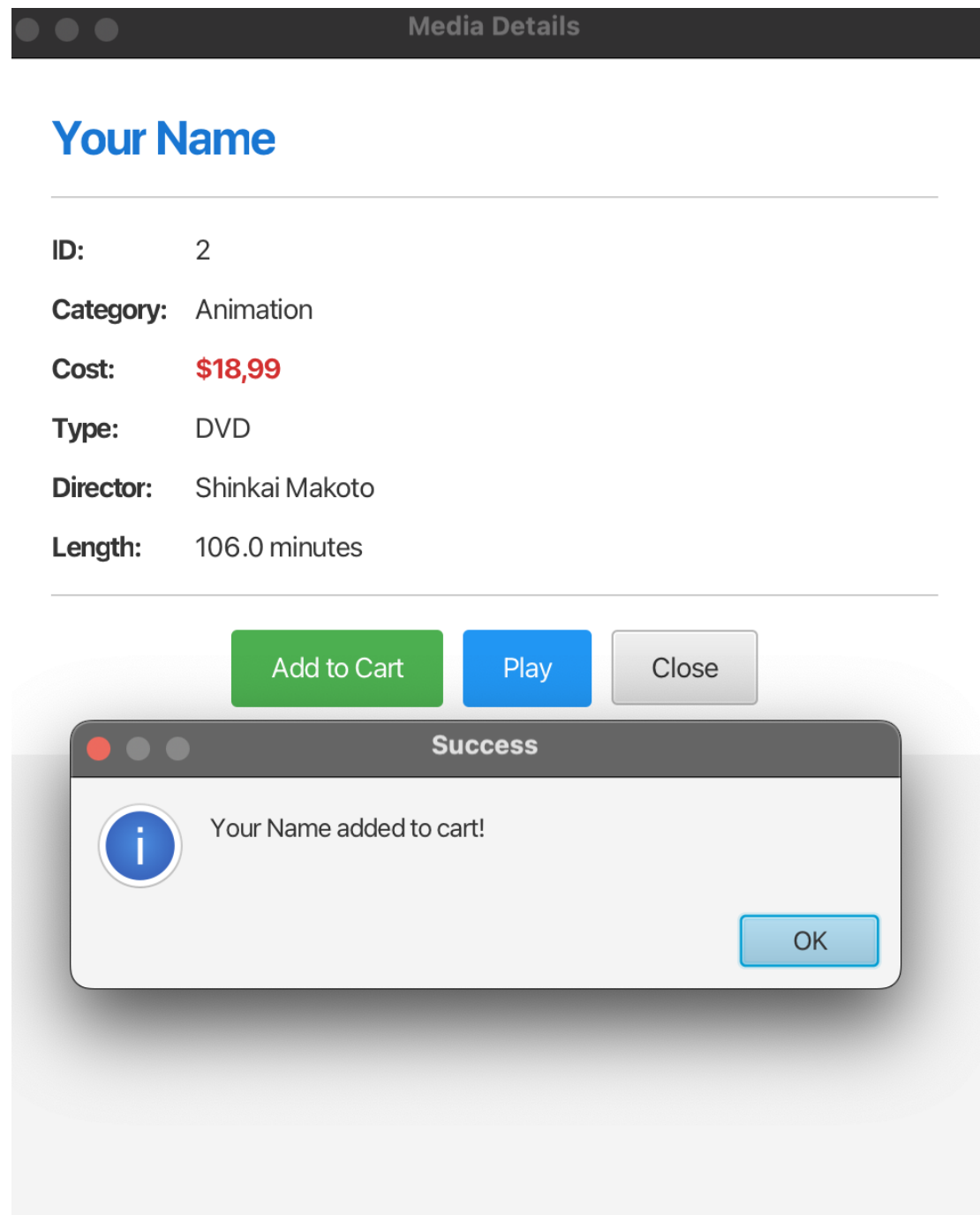


Hình 5: Play function

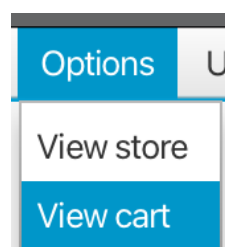
3.5 View Cart

3.5.1 Overview about the cart screen

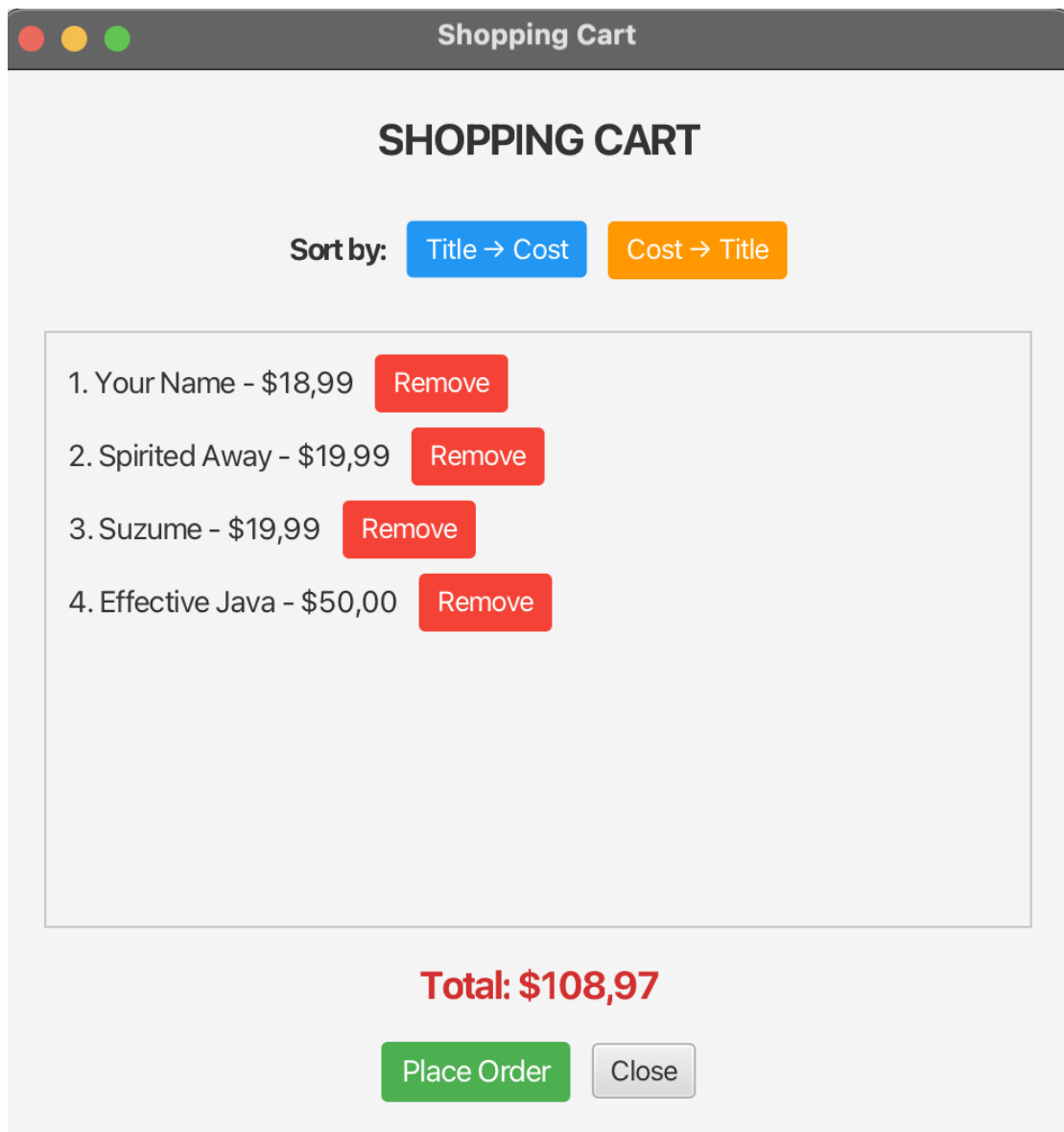
In the top left corner, We click to **Options** and choose **view Cart**



Hình 6: Add Your Name to the cart

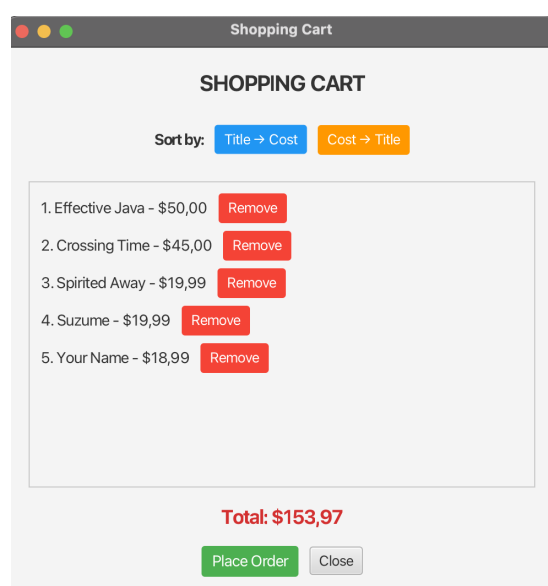
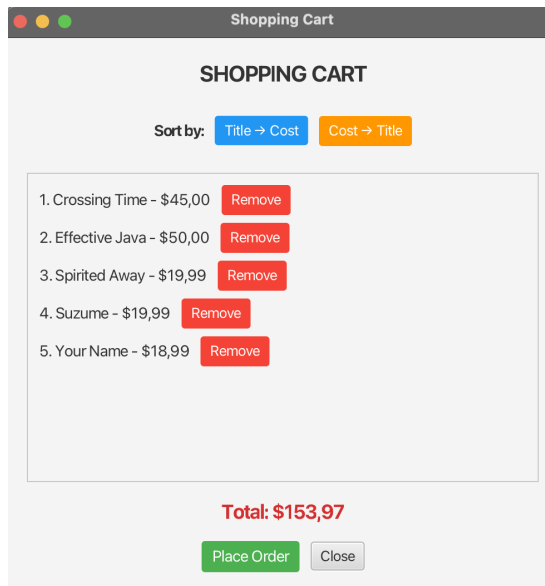


The Shopping Cart screen displays all added media items and their corresponding



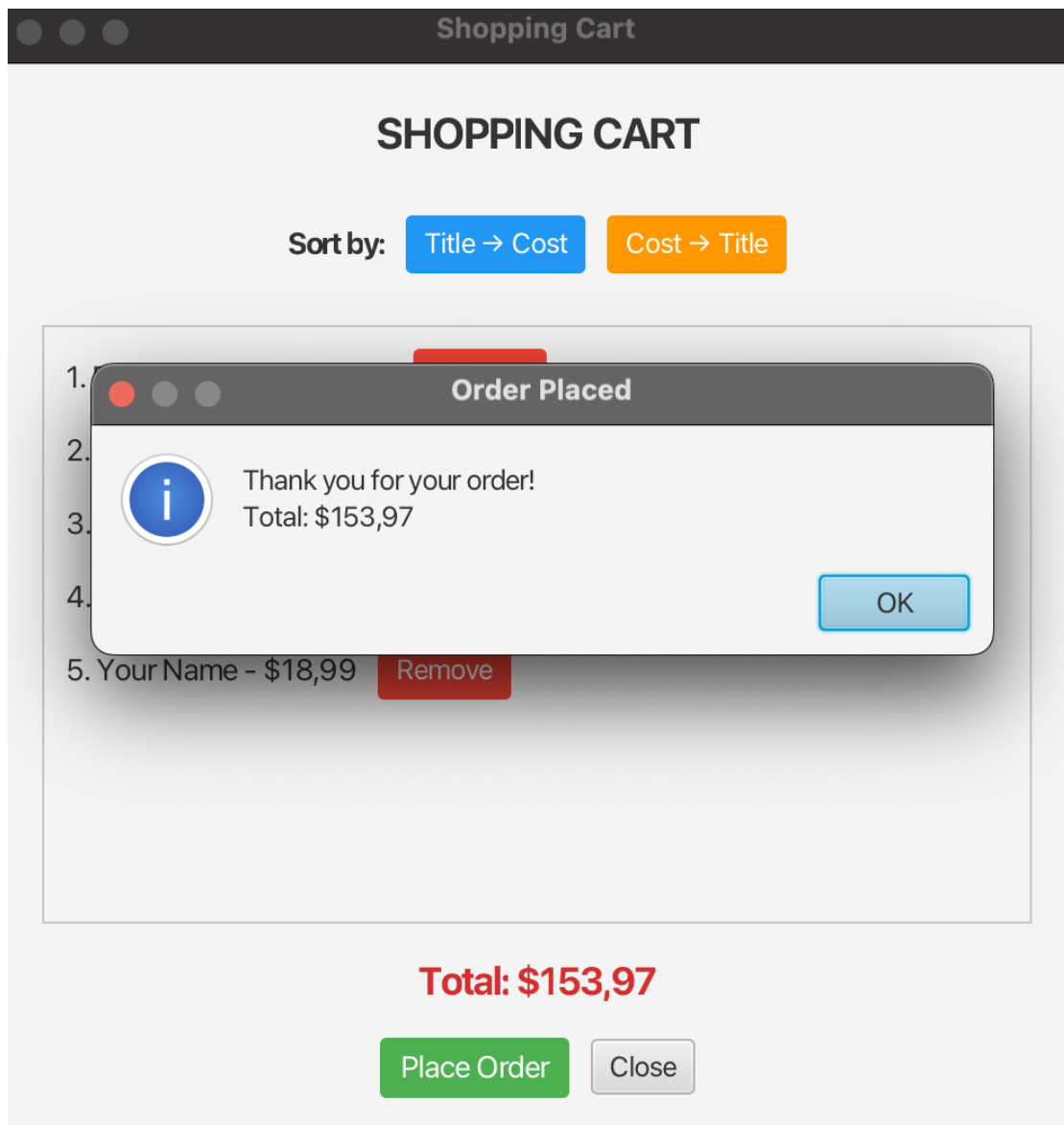
total price. Users have the options to remove any media from the cart, sort the items (by Title and then Cost, or vice versa), and finalize the transaction by clicking the **Place Order** button.

3.5.2 Sort



Hình 7: Sort the media by Title → Cost Hình 8: Sort the media by Cost → Title

3.5.3 Place Order



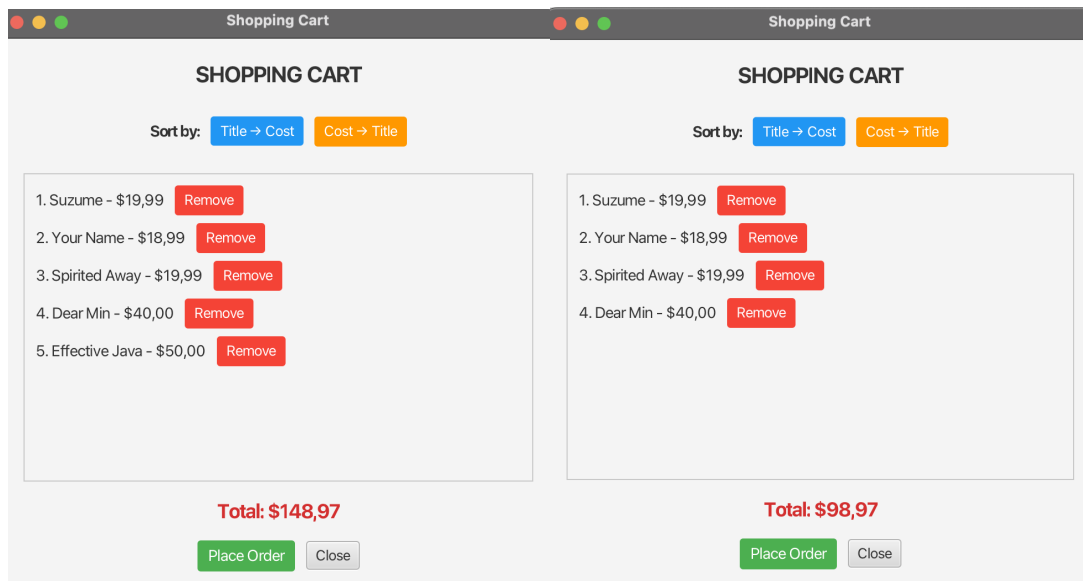
Hình 9: Place order

3.5.4 Remove

3.6 Store

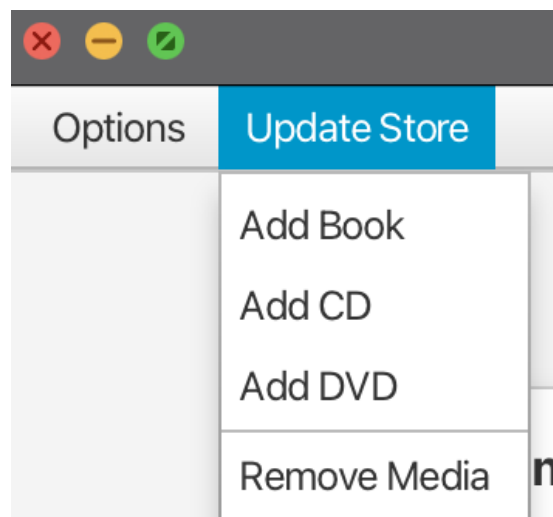
3.6.1 Update Store function

Beside the Options, we choose **Update Store** button



Hình 10: The cart before removal

Hình 11: The cart after removal

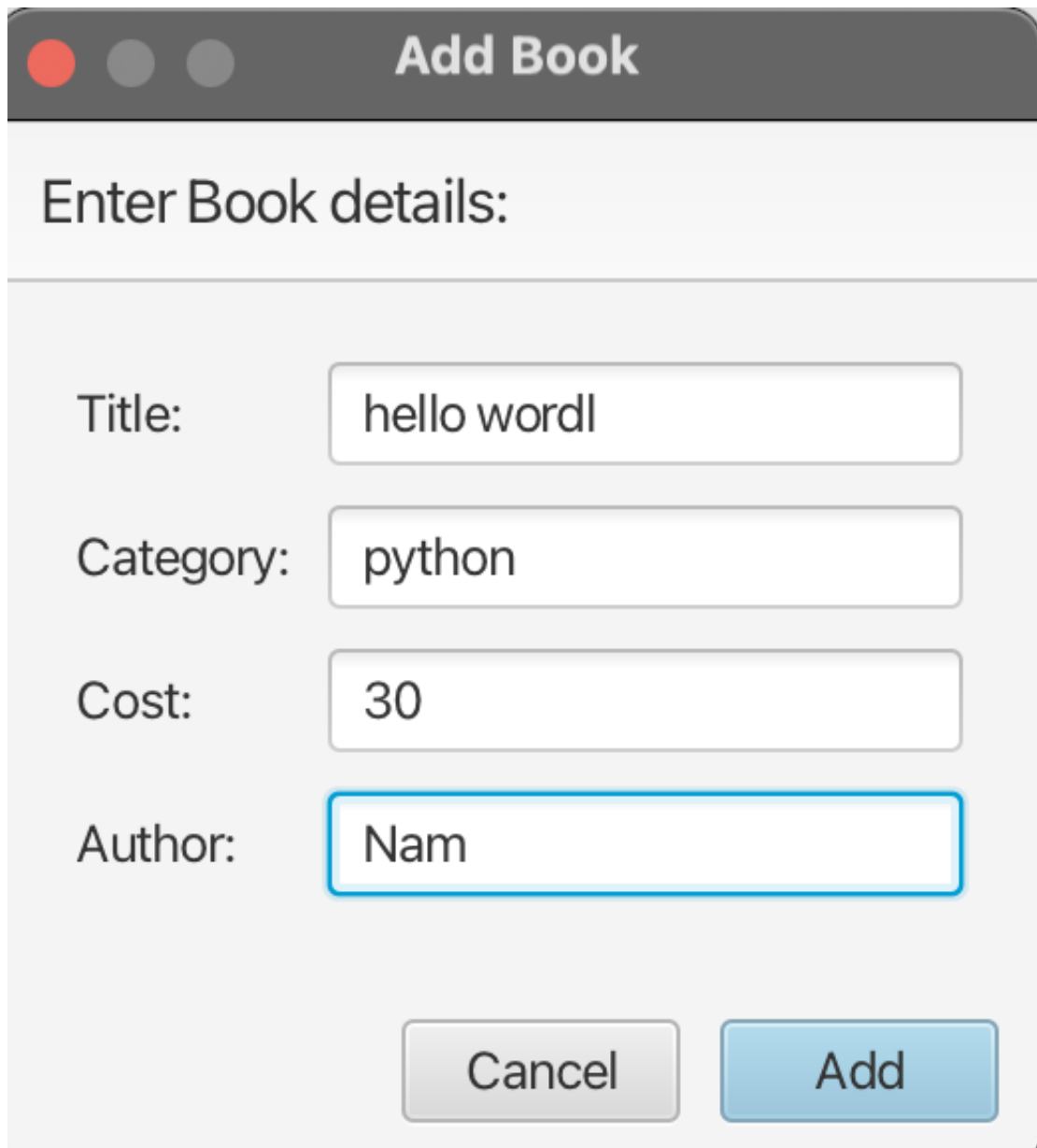


3.6.2 Add Book

The owner will enter the title, category, cost and author's name, and click button **Add** to finish **adding step**

3.6.3 Add Compact Disc

The owner will enter the title, category, cost and author's name, and number of tracks to generate the field corresponding to the number. After that, the owner will enter the title and length of each track.



Add Book

Enter Book details:

Title:

Category:

Cost:

Author:

Hình 12: Add new book Hello World

Add Compact Disc

CD Information

Title:

Category:

Artist:

Cost:

Tracks

Number of tracks:

Track 1

Title: Length:

Track 2

Title: Length:

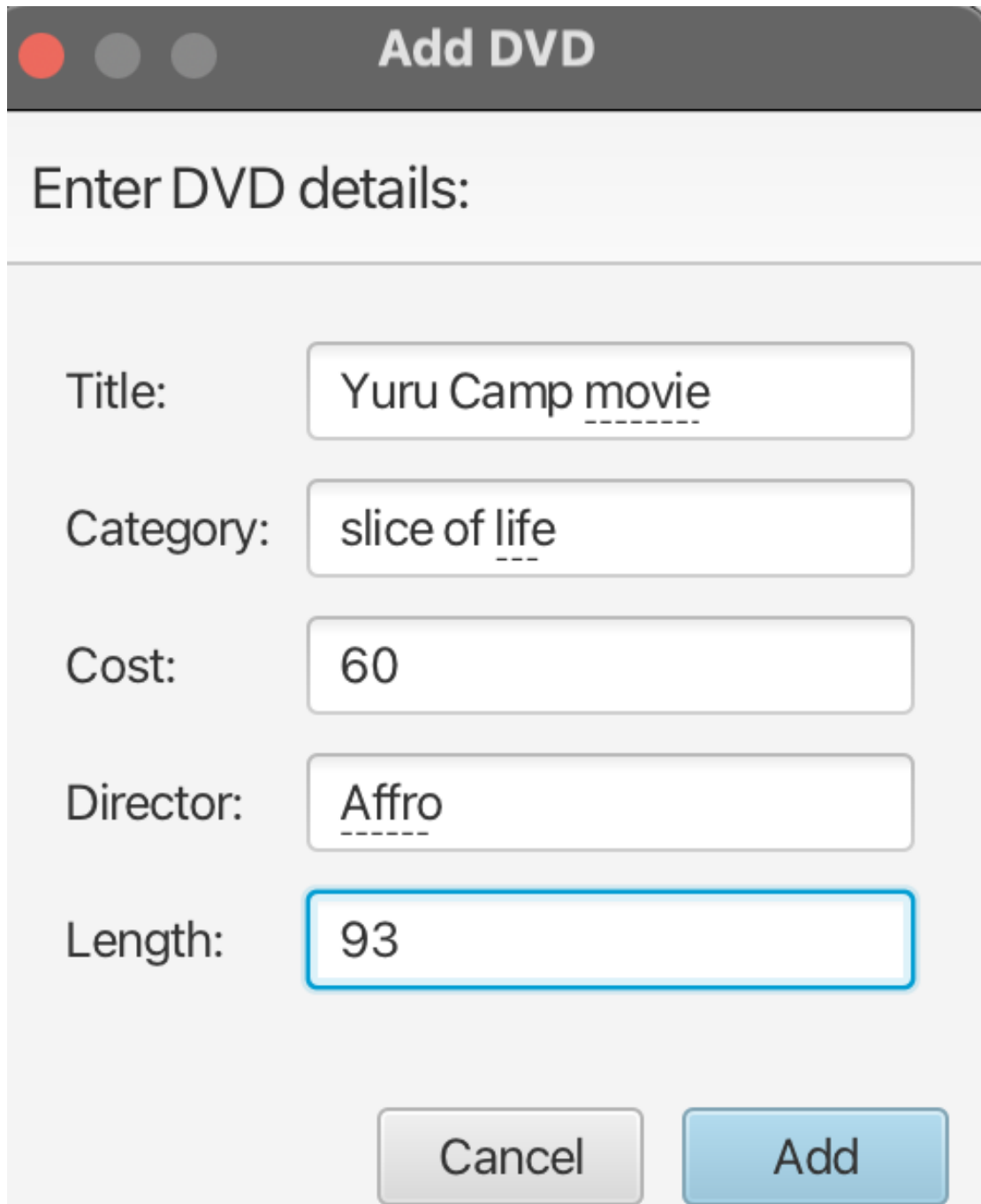
Track 3

Title: Length:

Hình 13: Add new CD Đánh đổi

3.6.4 Add DVD

The owner will enter the title, category, cost and Director's name, and length(must be positive), if not, the screen will show the exception. This part we will mention below.



Add DVD

Enter DVD details:

Title: Yuru Camp movie

Category: slice of life

Cost: 60

Director: Afro

Length: 93

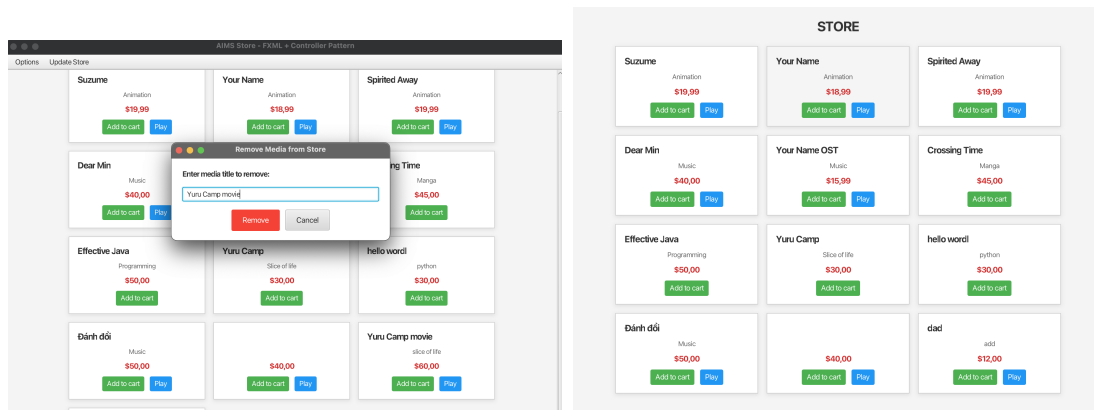
Cancel Add

Hình 14: Add new DVD

3.6.5 Remove media

4 Exception Handling

Exception handling allows us to manage basic error scenarios, informs the user about the specific nature of the fault, thereby facilitating maintenance, and prevents the system from crashing due to simple errors.



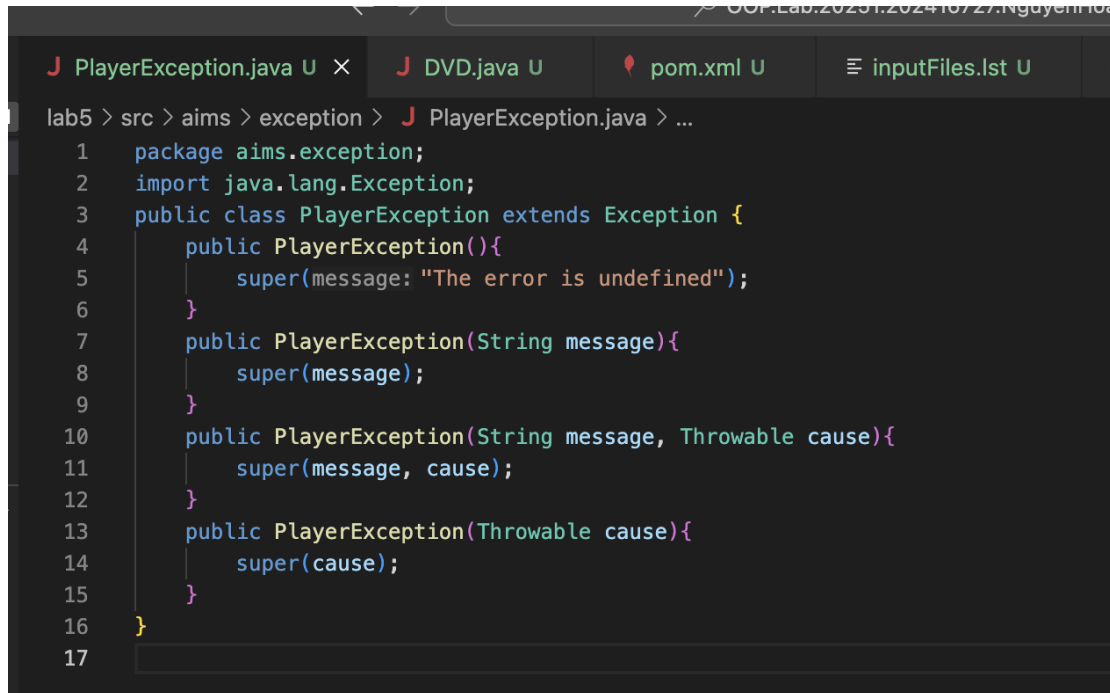
Hình 15: Remove YuruCamp Movie

Hình 16: YuruCamp Movie has been removed

Hình 17: *
Store after remove

Function	Reason	Action
Play Media	Media has length ≤ 0	Throw <code>PlayerException</code>
Add Media (Store)	Invalid input	Throw <code>InvalidInputException</code>
Add Media (Cart)	The limit reach MAXIMUM	Throw <code>LimitExceedException</code>

4.1 Exception for Play Media



Hình 18: PlayerException class

4.1.1 DVD + Tracks

When customer play demo DVD or Track which has the length non-positive, it will show the message error **"The DVD has the length less than or equal 0"**. So that the system will not crash and inform the owner that the DVD meet error, so he will fix this.



Hình 19: Play function of DVD

```

@Override
public void play() throws PlayerException {
    if (this.getLength() <= 0) {
        throw new PlayerException(message: "ERROR: Track length is non-positive!");
    }
    System.out.println("Playing track: " + title + " (" + length + " mins)");
}

```

Hình 20: Play function of Track

4.1.2 Compact Disc

```

53 @Override
54 public void play() throws PlayerException{
55     if (this.getLength() <= 0 ) {
56         throw new PlayerException(message: "ERROR: CD length is non-positive");
57     }
58
59     System.out.println("Playing CD: " + this.getTitle());
60     System.out.println("Artist: " + this.artist);
61     System.out.println("Total tracks: " + tracks.size());
62     System.out.println(x: "=====");
63
64     int successCount = 0;
65     int failCount = 0;
66     StringBuilder errors = new StringBuilder();
67
68     for (int i = 0; i < tracks.size(); i++) {
69         Track track = tracks.get(i);
70         System.out.print("Track " + (i + 1) + ": " + track.getTitle() + " - ");
71
72         try {
73             track.play();
74             successCount++;
75         } catch (PlayerException e) {
76             failCount++;
77             System.out.println("ERROR: " + e.getMessage());
78             errors.append(str: "\nTrack ").append(i + 1)
79                 .append(str: ": ").append(track.getTitle()).append(str: "\n")
80                 .append(str: "\nERROR: ").append(e.getMessage())
81                 .append(str: "\nLength: ").append(track.getLength()).append(str: " min");
82         }
83     }
84
85     System.out.println(x: "=====");
86     System.out.println("Playback summary: " + successCount + " succeeded, " + failCount + " failed");
87
88     if (failCount > 0) {
89         throw new PlayerException("There " + failCount + " are track(s) unplayable:" + errors.toString());
90     }
91 }

```

Hình 21: Play function of Compact Disc

The system is designed to throw a **PlayerException** if an individual track's length is less than or equal to zero. However, the core requirement is to prevent the entire media player from stopping when a single track fails; instead, the system must attempt to play all subsequent tracks. To achieve this fault tolerance and provide comprehensive user feedback, nested try-catch blocks are implemented.

4.2 Exception for Limitation

```
1 package aims.exception;
2
3 public class LimitExceededException extends Exception {
4     public LimitExceededException() {
5         super();
6     }
7
8     public LimitExceededException(String message) {
9         super(message);
10    }
11
12    public LimitExceededException(String message, Throwable cause) {
13        super(message, cause);
14    }
15
16    public LimitExceededException(Throwable cause) {
17        super(cause);
18    }
19 }
20
```

Hình 22: LimitException

```
public void addMedia(Media media) throws LimitExceededException {
    if (itemsOrdered.size() >= MAX_NUMBERS_ORDERED) {
        throw new LimitExceededException("ERROR: The number of media has reached its limit (" + MAX_NUMBERS_ORDERED + " items)");
    }

    if (!itemsOrdered.contains(media)) {
        itemsOrdered.add(media);
        System.out.println("The media " + media.getTitle() + " has been added!");
    } else {
        System.out.println("The media " + media.getTitle() + " is already in the cart!");
    }
}
```

Hình 23: Cart Limit

When the cart reaches the **MAXIMUM_NUM_ORDER**, the system will display the ERROR message: **"The number of media in the cart has reached the limit"**.